

AMENDMENT NO. 2 AUGUST 1935
TO
IS:10150-1981 GUIDE FOR STERILIZATION OF
MEDICAL PRODUCTS

(Page 8, clause 4.2.1) - Add the following new clauses after 4.2.1 and renumber the subsequent clauses accordingly:

'4.3 Sterilization by Formaldehyde Gas – The material to be sterilized is exposed to formaldehyde gas at low temperature (approximately 55°C). The process is carried out in a specialised vessel. The process depends on four factors, namely, temperature, relative humidity, gas concentration and exposure time. Besides, the product design which includes the penetrability, the packaging system employed, bioburden of the product and the state of hydration of the micro-organisms is also important.

4.3.1 *Bioburden* - A bioburden history of the product shall be established since this factor influences the effectiveness of the sterilization process.

4.3.2 *Packaging System* - The packaging materials shall be highly permeable to formaldehyde gas or alternatively, a breathing system may be provided which allows the penetration of the formaldehyde gas into the product. The material such as double thickness muolin or paper or single thickness polyethylene (PE) or polypropylene, laminates as cello-PE, mylar-PE, foll-PE may be used.

4.3.3 *Sterilization Process* - Formaldehyde, gas sterilization requires a specialised equipment designed for this particular purpose. The instructions provided by the manufacturer of the formaldehyde gas sterilizer shall be followed. The following exposure conditions shall be achieved for the formaldehyde gas sterilization:

- Temperature - $55^{\circ}\text{C} \pm 2^{\circ}\text{C}$
- Vacuum - 700 mm $\pm$ 20 mm before gas entry
in the chamber and
600 mm $\pm$ 20 mm during the
sterilization process
- Relative humidity - 55 to 60 percent
- Exposure time - 10 - 12 hours
- Gas concentration in the chamber - 400 mg - 500 mg
per litre

The above cycle shall be monitored by a recording system which may be manual or automatic.

4.3.4 *Biological Indicator* - The biological indicators are required for the purpose of determination of sterility. Biological indicators shall be used to monitor effectiveness of the sterilization of the product. At least 10 spore strips shall be used of 10^6 *Bacillus subtilis* var *niger* (NCTC 10073) for each sterilising cycle.'